

Homework 12: Soil, Impressions & Advanced Prints

Paper track. Pairs with Home Lab 12. A soil texture triangle helps.

Part A: Soil texture triangle (2 pts each)

Name the texture class:

1. 70% sand, 20% silt, 10% clay
2. 15% sand, 65% silt, 20% clay
3. 33% sand, 33% silt, 34% clay
4. 45% sand, 40% silt, 15% clay
5. 10% sand, 30% silt, 60% clay

Part B: Soil facts (1 pt each)

1. Particle-size range for sand, silt, and clay.
2. What causes soil to be red? Dark brown/black? Gray?
3. Is soil class evidence or individual evidence?
4. A soil is 100% minus 55% sand minus 30% silt — what's the clay %? Then name the class.

Part C: Fingerprints (2 pts each)

1. List the eight IAFIS patterns.
2. Delta count for arch, loop, whorl.
3. A print shows two loops swirled together like a yin-yang. Which pattern?
4. Latent, patent, plastic — define each, and give one way to develop a latent.
5. Define: dermatoglyphics, dactyloscopy, adermatoglyphia, hexadactyly.
6. What does each stand for: AFIS, IAFIS, CODIS, NCIC? Which one is a DNA database?

Part D: Shoeprint (3 pts)

A crime-scene shoeprint and a suspect's shoe both show a men's size 10, a herringbone tread, and heavy wear on the outer heel. The crime-scene print also shows a **triangular gouge across the third tread bar**, which is present on the suspect's actual sole.

1. Which features are class characteristics? Which is an individual characteristic?
2. Does the size + tread + wear match prove it's the same shoe? Does the gouge?

Answer Key

Part A: 1. Sandy loam. 2. Silt loam. 3. Clay loam. 4. Loam. 5. Clay.

Part B:

1. Sand 2.0–0.05 mm; silt 0.05–0.002 mm; clay < 0.002 mm.
2. Red = iron oxides; dark brown/black = organic matter (humus); gray/blue = waterlogged / oxygen-poor.
3. Class evidence.
4. Clay = 15%; 55/30/15 → **sandy loam**.

Part C:

1. Plain arch, tented arch, radial loop, ulnar loop, plain whorl, central pocket loop whorl, double loop whorl, accidental whorl.
2. Arch 0, loop 1, whorl 2.
3. Double loop whorl.
4. Latent = invisible (sweat/oil) — develop with dusting powder, superglue fuming, ninhydrin, iodine fuming, or fluorescent dye. Patent = visible, transferred in blood/ink/grease/dust. Plastic = a 3-D impression in a soft material.
5. Dermatoglyphics = study of friction-ridge skin patterns; dactyloscopy = fingerprint identification; adermatoglyphia = born with no fingerprints; hexadactyly = born with a sixth finger or toe.
6. AFIS = Automated Fingerprint Identification System; IAFIS = Integrated AFIS; CODIS = Combined DNA Index System; NCIC = National Crime Information Center. **CODIS is the DNA database.**

Part D:

1. Class: size 10, herringbone tread, outer-heel wear. Individual: the triangular gouge across the third tread bar.
2. The size/tread/wear match is class evidence — many shoes fit, so it doesn't prove it. The gouge is an individual characteristic — it identifies this one specific shoe.